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Limits to generality: seasonal and temporal variation in dispersal of an intertidal gastropod

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Abstract

Seasonal and temporal variation can prevent the generalization of ecological models to situations other than those in which they were developed. This paper reports a study specifically designed to test the models that seasonal or temporal variation prevents the accurate prediction of dispersal of intertidal gastropods (*Bembicium auratum*) in mangrove forests using models developed on rocky shores. Analysis of variance was used to compare directly the outcomes of experiments replicated several times and in different seasons. In general, the dispersal of *Bembicium* was very consistent through time, particularly on rocky shores. Differences in the behaviour of *Bembicium* inhabiting rocky shores and those inhabiting mangrove forests occurred in all runs of the dispersal experiment. The model that seasonal variations in the dispersal of *Bembicium* inhabiting one habitat prevented prediction of dispersal in the other was rejected. There was, however, some aseasonal temporal variation in the dispersal of *Bembicium*, particularly at one of the mangrove forests. This suggests that prediction of dispersal in mangrove forests will be less precise than for rocky shores unless further work is done to evaluate causes of temporal variation in that habitat. Experiments must be replicated both spatially and temporally in order to assess the generality of ecological processes, to facilitate proper comparisons with other studies and to provide a sounder basis for the prediction of ecological processes within and between habitats. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Gastropod; Dispersal; Seasonal variation; Temporal variation; Generality; Meta-analysis

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1. Introduction

An important aim of ecology is to compare processes over broad spatial and temporal domains in search of general, predictable patterns (Underwood and Fairweather, 1986; Tilman, 1989; Peters, 1991; Underwood and Petraitis, 1993). This approach is not without its difficulties, however. In particular, there are many sources of variation that can confound comparisons between different regions or prevent the use of general models to predict processes in previously unstudied habitats (Underwood and Denley, 1984; Underwood and Petraitis, 1993; Beck, 1997; Crowe, in review). If we are to develop useful general models, it is important to design research programs to test specific hypotheses about generalizations among habitats and to investigate processes that limit generalization.

Temporal changes in underlying sources of variation present a classic example of this. Although some processes are remarkably consistent (e.g. Underwood and Chapman, 1992), or vary with predictable seasonality (e.g. Vadas, 1992), many ecological processes vary on time scales with no clearly apparent pattern (e.g. Wiens, 1986; Morrisey et al., 1992; Kennelly and Underwood, 1993; Haynes and Quinn, 1995). Irregular variations through time can make the prediction of ecological patterns very imprecise. Confident predictions can only be made if there is some evidence that the process will be consistent from one time to the next. It is impossible to judge a priori whether a given pattern or process will be variable or consistent through time. Consequently, several authors have recognised the need for independent repetitions of ecological studies to assess temporal generality (e.g. Connell, 1974 (cited by Elner and Vadas, 1990); Underwood and Chapman, 1992).

Here experiments involving the littorinid snail *Bembicium auratum* (Quoy and Gaimard) were done several times to test whether the results of repeated experiments would vary through time and thus affect predictions both within and among habitats. *Bembicium* inhabits estuarine rocky shores and mangrove forests in southern Australia (Reid, 1988). It maintains an association with microhabitat provided by oysters (Crowe and Underwood, 1997). Crowe (1996a) investigated the dispersal of this species from patches of microhabitat on rocky shores near Sydney and experimentally tested whether results from rocky shores could be used to predict dispersal of *Bembicium* in mangrove forests. There was less dispersal in mangrove forests than on rocky shores. In addition, there was no effect of spatial heterogeneity of microhabitat on patterns of dispersal in mangroves, whereas, on rocky shores, juveniles sometimes dispersed less from patches of oysters that were isolated from surrounding oysters, than from patches contiguous with other oysters. These differences limit the generalization of models from rocky shores to mangrove forests. One possible explanation for the differences has already been discounted: they are not due to intrinsic differences in the snails inhabiting the habitats. Instead, the differences are caused by responses of the snails to extrinsic cues which differ from one habitat to the next (Crowe and Underwood, in review). The current paper examines further models to account for the results presented in Crowe (1996a) and builds on the findings of Crowe and Underwood (in review). The aim is to identify processes that can limit the generality of ecological models and to document temporal variation in dispersal of this species, thus enabling proper comparisons to be made with other appropriately designed studies (see Underwood and Petraitis, 1993).

One model that could explain the findings to date is that extrinsic behavioural cues vary seasonally in different ways from one habitat to the other. Seasonal variations in extrinsic factors have often been shown to affect ecological processes in general (e.g. Underwood, 1975; Dean and Hurd, 1980; Frank, 1982; Underwood and Jernakoff, 1984; Haynes and Quinn, 1995; Recher et al., 1996; Anderson, 1995) and the behaviour of gastropods in particular (e.g. Castenholz, 1961; Feare, 1971; Bertness et al., 1983; Vadas, 1992; Williams, 1993). Underwood and Barrett (1990) investigated movements of *Bembicium* in mangrove forests at Woolooware Bay. An experiment was repeated three times in autumn and once in summer. Although *Bembicium* moved during the autumnal repetitions of the experiment, they did not move at all in the summer. The first run of the dispersal experiment described by Crowe (1996a) was done in the summer (Crowe, 1996b). There was less movement in mangrove forests than on rocky shores. This difference could be explained if cues in mangrove forests stimulate snails to move less in summer than at other times of the year, as suggested by the results of Underwood and Barrett (1990). If this model were true, the behavioural differences between the habitats should be apparent only in the summer. Replicate sets of experimental data were collected in summer and in winter to test this hypothesis.

2. Methods

2.1. Experimental procedures

Subsets of published data from two different types of experiment were used to make the relevant comparisons. These experiments were described in detail in Crowe (1996a) and Crowe and Underwood (in review) respectively and are outlined only briefly here.

The experiments were done at two locations on estuarine rocky shores and two locations in mangrove forests to the north and south of Sydney. The rocky shores were in Port Hacking and Broken Bay. The mangrove forests, in Woolooware Bay and Patonga Creek, were geographically interspersed among the rocky locations (see Crowe, 1996a for further details).

Standard plots of oysters were constructed at these locations in order to test the hypotheses. All plots were 15×15 cm. To test hypotheses about the effects of the spatial heterogeneity of microhabitat, plots were either contiguous with surrounding oysters or isolated from them by a halo of primary substratum: rock or mud and pneumatophores, depending on the habitat.

In experiments of the first type, called dispersal experiments, marked juvenile and adult *Bembicium* were placed on replicate contiguous and isolated plots at each of the locations. For the purposes of the experiments, *Bembicium* ≥ 10 mm were considered to be adults and animals ≤ 8 mm were juveniles (see Crowe, 1996a for rationale). They were left for two days to disperse and then the following data were collected: the numbers of snails remaining on the plots, the numbers of snails no longer on the plots and the distances by which snails had become displaced from the plots. Any snail that remained on a plot was assigned a displaced distance of 0 cm.

The second type of experiment involved reciprocal experimental transplantations designed to test the effects on the dispersal of *Bembicium* of moving them between

habitats. One of the controls was exactly the same as the experimental treatment in the dispersal experiment described above, and data from these plots were used here.

2.2. Seasonal and temporal comparisons

Three analyses were done for each comparison. The first was of the percentages of snails that had dispersed from plots after two days and the other two were of their displaced distances. Measurements of individual distances were not analysed because the analyses already involved five factors and would have become unnecessarily complex. Instead, separate analyses were done of the means and variances of displaced distances from the plots.

In the replicate runs of the experiments, there were variable numbers of replicate plots (Table 1). In order to balance designs for the analyses of seasonal and temporal variation, *n* was chosen as the minimal number of plots available from any one replicate run in the comparison. Data from other replicate runs were discarded at random to balance the samples.

For the comparison of dispersal in different seasons, 'Summer' was represented by the first run of the dispersal experiment (Summer 1993) and the transplantation experiment (Summer 1995). 'Winter' was represented by the second and third runs of the dispersal experiment (Winter 1994 and Winter 1995). The summer and winter replicates were thus interspersed through time (summer, winter, summer, winter). The use of data from the transplantation experiment imposed two limitations on the design: (a) the experiment used only adult snails, so it was not possible to include Age as a factor in the seasonal comparison and (b) only one of the five original treatments ('TL3', see Crowe and Underwood, in review) provided data comparable to that yielded by the dispersal experiments and displaced distances were measured for only two of the three plots assigned to that treatment. Therefore, only three and two replicate plots per cell were available for analyses of percentage dispersal and displaced distance respectively (Table 1).

In order to balance the design for the seasonal comparison of percentage dispersal, three replicate plots were selected from those available for each of the dispersal experiments. This resulted in the loss of at least half the available data from each of the Summer 1993 and Winter 1994 runs. It is conceivable that a subset comprising only half of the available data could give a misleading representation of the data set as a whole. If

Table 1
Data available for temporal and seasonal comparisons

Experiment	Repetition	Age	Plots available
Dispersal	Summer (Oct - Dec) 1993	J and A	8
	Winter (Jun - Sep) 1994	J and A	6
	Winter (Jul - Aug) 1995	J and A	4
Transplant	Summer (Feb) 1995	A only	3 [a]

The last column gives the number of comparable replicate plots per location in each cell of the design. Subsets of data were randomly selected from this number of plots (see text for details). J = juveniles, A = adults.
[a] Displaced distances only measured for 2 plots (Crowe and Underwood, in review).

this were the case, a comparison of two randomly selected subsets from within each of those runs in a nested design would yield significant differences among subsets or interactions involving the term 'Subset'. An additional analysis was done to test this hypothesis.

Temporal variation was assessed by comparing three replicate runs of the dispersal experiment (Table 1). Data were available for both age classes of snails, enabling the inclusion of Age as a factor in the analysis. The minimal number of replicates was 4 (in Winter 1995).

Strictly speaking, plots were not independent in the different runs of the experiments, in the sense that the same set of plots were used throughout. It would have been too destructive to construct a new set of plots for each repetition, indeed there would not have been sufficient space at the locations on rocky shores. The plots were, however, randomly re-assigned to treatments for each repetition and there were at least 5 months between replicate runs.

Where appropriate, terms that were not significantly different ($P > 0.25$) were pooled to enable tests to be made of previously untestable terms and to provide more powerful tests of additional terms ('post-hoc pooling': Winer, 1971; Underwood, 1981). In general, however, the use of pooling was kept to a minimum. Only terms of relevance to the hypotheses being tested were re-evaluated after pooling.

2.3. Comparisons with controls

There were no differences in how the snails on the different types of plot were handled, so comparisons could be made between snails of different ages on plots of different types without reference to controls. Controls for the effects of handling the snails were necessary, however, to assess the relevance of the results to natural populations of unmanipulated snails (Chapman, 1986; Underwood, 1986, Underwood, 1988). Details of procedures and analyses of variance comparing control and experimental treatments can be found in Crowe (1996a), Crowe (1996b) and Crowe and Underwood (in review). There were no consistent differences among control and experimental treatments at any time for any combination of Age, Isolation, Location and/or Habitat.

3. Results

3.1. Seasonal variation

There was no evidence that the random selection of data had provided misleading estimates of dispersal: no interactions involving the term 'Subset' were significant (Table 2). In addition, when subsets of data were compared with the mean of the complete set of data, only 8 subsets (out of 32) had a mean that was different by 5% or more from that of all the data available. Of those, 3 were different by less than 10 percentage points and 5 were different by 15 or more percentage points. Seven of the eight cases in which the subset was noticeably different from the complete set were from

Table 2
Analysis comparing percentage dispersal of adult *Bembicium* from two randomly selected subsets of data from each of two temporal repetitions of the dispersal experiment

Source of variation	df	MS	*F*
Time = T	1	459.37	No test
Subset = Su(T)	2	176.04	0.36
Habitat = H	1	36426.04	No test
Location = L(H)	2	2042.71	1.22
Isolation = I	1	651.04	No test
TxH	1	26.04	No test
TxL(H)	2	1667.71	3.41
TxI	1	126.04	No test
HxSu(T) [a]	2	67.71	0.14
L(H)xSu(T) [b]	4	488.54	1.01
IxSu(T) [c]	2	113.54	0.40
HxI	1	551.04	No test
IxL(H) [d]	2	192.71	0.40
TxHxI	1	176.04	No test
IxTxL(H) [e]	2	476.04	1.67
IxHxSu(T)	2	755.21	[1]1.55
IxL(H)xSu(T) [f]	4	284.37	0.59
Residual [g]	64	483.33	

[1] *F* calculated using pooled MS (a + b + c + d + e + f + g = 484.29 with 76 df).
($n = 3$ plots; data are untransformed; Cochran's test: ns). Experimental treatments only. Initial densities were 5 juveniles and 5 adults per plot in mangrove forests and 10 juveniles and 10 adults per plot on rocky shores.

mangrove forests. This observation could be explained by greater variation from plot to plot in mangrove forests than on rocky shores. To test this, the variance among plots was compared in the two habitats. Each location contributed one replicate measurement of variation among plots to the analysis. The variance among plots in mangrove forests (mean = 546.6) was consistently greater than that on rocky shores (mean = 171.2; Fig. 1; Table 3, $P < 0.01$).

On rocky shores, there was no seasonal variation in patterns of dispersal. Dispersal varied only slightly around a mean of approximately 75% at both locations in both seasons (Fig. 2a). In mangrove forests, on the other hand, there was some variation through time (Fig. 2b). The patterns in relation to season were, however, not consistent at the two locations (Isolation × Season × Location(Habitat), $P < 0.05$, Table 4a). Although there was a trend for *reduced* dispersal at Woolooware in winter compared to summer, the difference was not statistically significant (SNK procedure $P > 0.05$). Dispersal at Patonga was, by contrast, significantly *greater* in the winter than in the summer (SNK procedure $P < 0.05$). Dispersal in winter at Patonga was comparable to that on rocky shores (~75%), whereas in summer at Patonga and in both seasons at Woolooware, dispersal was generally less than 30% (Fig. 2b).

Displaced distances were consistently greater on rocky shores than in mangrove forests. In this respect, the behaviour of snails in the different habitats did not converge during the winter months (Fig. 3). The analysis was unambiguous. The interaction between Season, Habitat and Isolation was not significant. Although the interaction

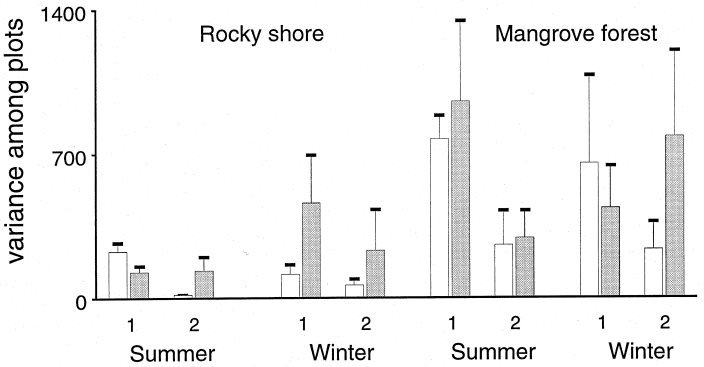

Fig. 1. Seasonal variation in among-plot variation of percentage dispersal of adult *Bembicium* in rocky shores and mangrove forests. Summer 1 = Dispersal, 1993; Summer 2 = Transplantation, 1995; Winter 1 = Dispersal, 1994; Winter 2 = Dispersal, 1995. Mean values for among-plot variation in percentage dispersal for each habitat + SE (n = number of locations = 2) are shown. Initial numbers per plot were: 10 adults and 10 juveniles on rocky shores and 5 adults and 5 juveniles in mangrove forests. □ = plots contiguous with surrounding oysters; ▨ = plots isolated from surrounding oysters.

between Season and Habitat could not be tested, even after pooling, its Mean Square was smaller than the error Mean Square and was clearly not a major source of variation (Table 4b). There was, however, temporal variation that was unrelated to season, particularly in mangrove forests (Fig. 3). At Woolooware, snails moved greater distances on average from contiguous plots in Summer 1995 than in Summer 1993, but the reverse was true of isolated plots. At Patonga, there was greater overall movement in Summer 1995 than in Summer 1993 and also in Winter 1995 compared to Winter 1994. There was also some spatial variation within any given time among locations on rocky shores

Table 3
Analysis of seasonal variation in variances among plots of percentage dispersal of adult *Bembicium*

Source of variation	df	MS	*F*
Season = S	1	4809.40	0.02
Time = T(S)	2	255007.96	[1]3.08
Habitat = H	1	1127531.16	[1]13.61**
Isolation = I	1	143909.71	3.24
SxH [a]	1	34897.24	0.28
SxI [b]	1	46313.04	1.04
HxT(S) [c]	2	124956.77	1.37
IxT(S) [d]	2	44457.60	0.49
HxI [e]	1	40.39	0.00
SxHxI [f]	1	18016.09	0.14
IxHxT(S) [g]	2	128916.14	1.42
Residual [h]	16	91085.58	

** denotes significance at $P < 0.01$.
(n = 3 plots; data are untransformed; Cochran's test: ns). Experimental treatments only. Initial densities were 5 juveniles and 5 adults per plot in mangrove forests and 10 juveniles and 10 adults per plot on rocky shores.
[1]F calculated using pooled MS (a + b + c + d + e + f + g + h = 82819.11).

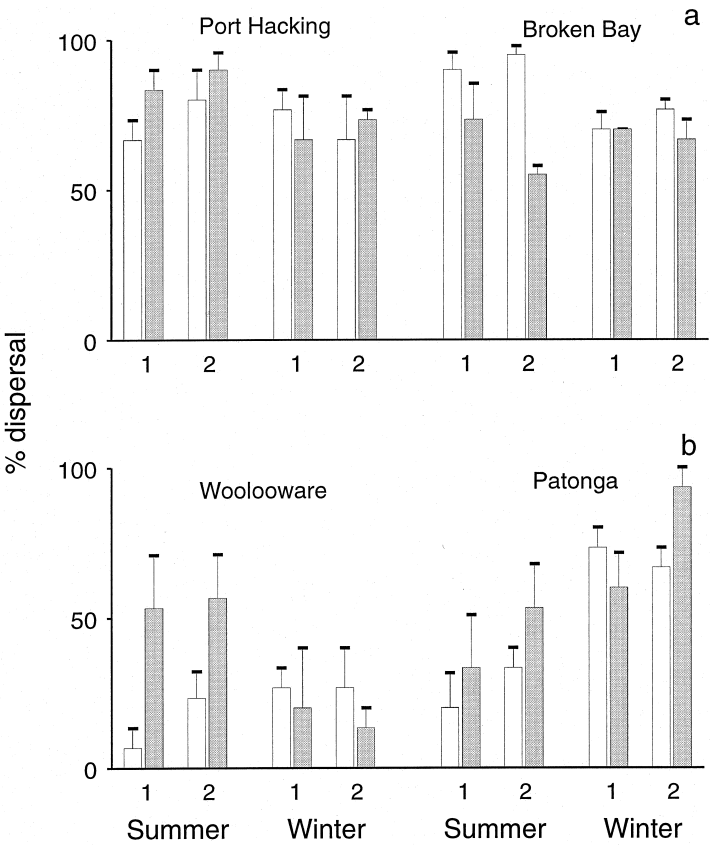

Fig. 2. Seasonal variation in percentage dispersal of adult *Bembicium*. a. rocky shores; b. mangrove forests. Summer 1 = Dispersal, 1993; Summer 2 = Transplantation, 1995; Winter 1 = Dispersal, 1994; Winter 2 = Dispersal, 1995. Mean percentages of initial numbers dispersed + SE (n = number of plots = 3) are shown. Initial numbers were: 10 adults and 10 juveniles on rocky shores and 5 adults and 5 juveniles in mangrove forests. □ = plots contiguous with surrounding oysters; ▨ = plots isolated from surrounding oysters.

and in mangrove forests. For example, in the Summer of 1995, snails at Port Hacking moved greater distances from contiguous plots than snails at Broken Bay and in Winter 1995, dispersal at Patonga was greater than that at Woolooware from both contiguous and isolated plots ($I \times L(H) \times T(S)$, $P < 0.01$, Table 4b; SNK procedure, $P < 0.05$).

As further evidence of the lack of a clear seasonal pattern in the behaviour of this species in either habitat, there were no significant seasonal differences in variances within plots (Table 4c).

3.2. Temporal variation

Broadly, the pattern across three replicate runs of the dispersal experiment was remarkably consistent from one time to another (Fig. 4). Dispersal was consistently

Table 4
Analyses of seasonal variation in dispersal of adult *Bembicium*. (a) Percentage dispersal ($n = 3$ plots; data are untransformed; Cochran's test: ns); (b) mean displaced distances ($n = 2$ plots; data are transformed $X' = \mathrm{Log}_e(X + 1)$; Cochran's test: ns); (c) variance within plots of displaced distances ($n = 2$ plots; data are transformed $X' = \mathrm{Log}_e(X + 1)$; Cochran's test: ns). Experimental treatments only. Initial densities were 5 juveniles and 5 adults per plot in mangrove forests and 10 juveniles and 10 adults per plot on rocky shores

Source	(a)			(b)			(c)		
	df	MS	*F*	df	MS	*F*	df	MS	*F*
Season = S	1	104.17	No test	1	0.29	[4]0.12	1	0.14	No test
Time = T(S)	2	375.00	1.64	2	0.41	0.67	2	6.22	1.65
Habitat = H	1	27337.50	No test	1	57.50	No test	1	220.66	No test
Location = L(H)	2	4008.33	17.49*	2	1.90	3.14	2	14.71	3.90
Isolation = I	1	376.04	No test	1	0.02	No test	1	2.64	No test
S × H	1	2604.17	No test	1	0.14	No test	1	0.18	[6]0.01
S × L(H)	2	4008.33	17.49*	2	2.42	4.00	2	5.40	1.43
S × I	1	1001.04	No test	1	0.22	No test	1	4.01	No test
H × T(S)[a]	2	241.67	1.05	2	2.43	4.03	2	14.67	3.88
L(H) × T(S)[b]	4	229.17	0.75	4	0.60	2.20	4	3.78	1.99
I × T(S)[c]	2	276.04	0.83	2	0.33	0.25	2	1.34	1.47
H × I	1	2109.37	[1]2.72	1	2.37	No test	1	11.47	No test
I × L(H)	2	776.04	2.32	2	1.39	1.04	2	0.36	0.40
S × H × I	1	1751.04	[2]1.52	1	0.01	[5]0.01	1	0.02	[7]0.01
I × S × L(H)[d]	2	1151.04	[3]3.82*	2	0.98	0.74	2	1.94	2.12
I × H × T(S)[e]	2	117.71	0.35	2	2.29	1.72	2	1.47	1.61
I × L(H) × T(S)[f]	4	334.37	1.09	4	1.33	4.85**	4	0.91	0.48
Residual[g]	64	305.73		32	0.27		32	1.90	

*Denotes significance at $P < 0.05$; ** denotes significance at $P < 0.01$.
[1]tested over $MS_{I \times L(H)}$ after pooling; [2]tested over $MS_{I \times S \times L(H)}$ after pooling; [3]*F* calculated using pooled MS (a + b + c + e + f + g = 300.85 with 78 df); [4]tested over $MS_{S \times I}$ after pooling; [5]*F* calculated using pooled MS (d + e + f = 1.48); [6]tested over $MS_{H \times T}$ after pooling; [7]tested over $MS_{I \times S \times L(H)}$ after pooling.

greater on rocky shores than in mangrove forests and dispersal by adults was consistently greater than that by juveniles (H × I × A, $P < 0.05$, Table 5a; SNK procedure $P < 0.05$). The only effect of isolating plots from surrounding oysters was seen on rocky shores, where it resulted in decreased dispersal by juveniles compared with plots contiguous with surrounding oysters (SNK procedure, $P < 0.05$). For adults on rocky shores and for adults and juveniles in mangrove forests, there were no differences in dispersal from isolated and contiguous plots. This result is the same as that found in the experiment done in the Winter of 1994 (Crowe, 1996a).

The only location at which the percentage dispersal changed through time was Patonga. Dispersal was greater there in the Winter of 1995 than in either Summer 1993 or Winter 1994 (T × L(H), $P < 0.001$, Table 5a; SNK procedure, $P < 0.01$). In the Winter of 1995, dispersal at Patonga was greater than that at Woolooware (SNK procedure, $P < 0.01$). The increased rate of dispersal at Patonga in Winter 1995 was also apparent in the data on displaced distances. Snails were displaced by greater distances from isolated and from contiguous plots in that run of the experiment than in either of the previous two (Fig. 5; I × T × L(H), $P < 0.05$, Table 5b; SNK procedure, $P < 0.05$). Displaced distances for that run of the experiment were greater at Patonga than at

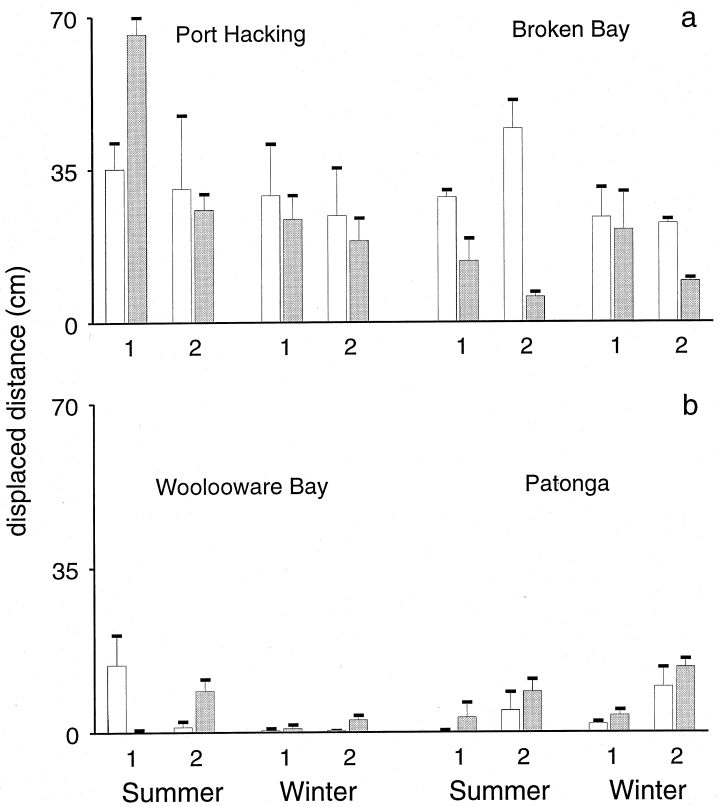

Fig. 3. Seasonal variation in displaced distances of adult *Bembicium*. a. rocky shores; b. mangrove forests. Summer 1 = Dispersal, 1993; Summer 2 = Transplantation, 1995; Winter 1 = Dispersal, 1994; Winter 2 = Dispersal, 1995. Mean displaced distances + SE (n = number of plots = 2) are shown. The number of individual measurements contributing to the mean for any given plot varied from 3 to 10 and was generally smaller in mangrove forests than on rocky shores. Initial numbers were: 10 adults and 10 juveniles on rocky shores and 5 adults and 5 juveniles in mangrove forests. □ = plots contiguous with surrounding oysters; ▨ = plots isolated from surrounding oysters.

Woolooware (SNK procedure, $P < 0.01$). An identical pattern emerged from the analysis of the variance within plots of displaced distances (I × T × L(H), $P < 0.05$, Table 5c; SNK procedure, $P < 0.01$). There was also greater variance in displaced distances from contiguous plots at Patonga than at Woolooware in Winter 1994. In both of the winter runs of the experiment, displaced distances at Patonga were more variable from isolated than from contiguous plots (I × T × L(H), Table 5c, $P < 0.05$; SNK procedure, $P < 0.05$).

There was also some temporal variation in displaced distances on rocky shores (Fig. 5). In the first run of the experiment, snails at Broken Bay moved greater (Table 5b) and more variable (Table 5c) distances from isolated plots than from contiguous plots and

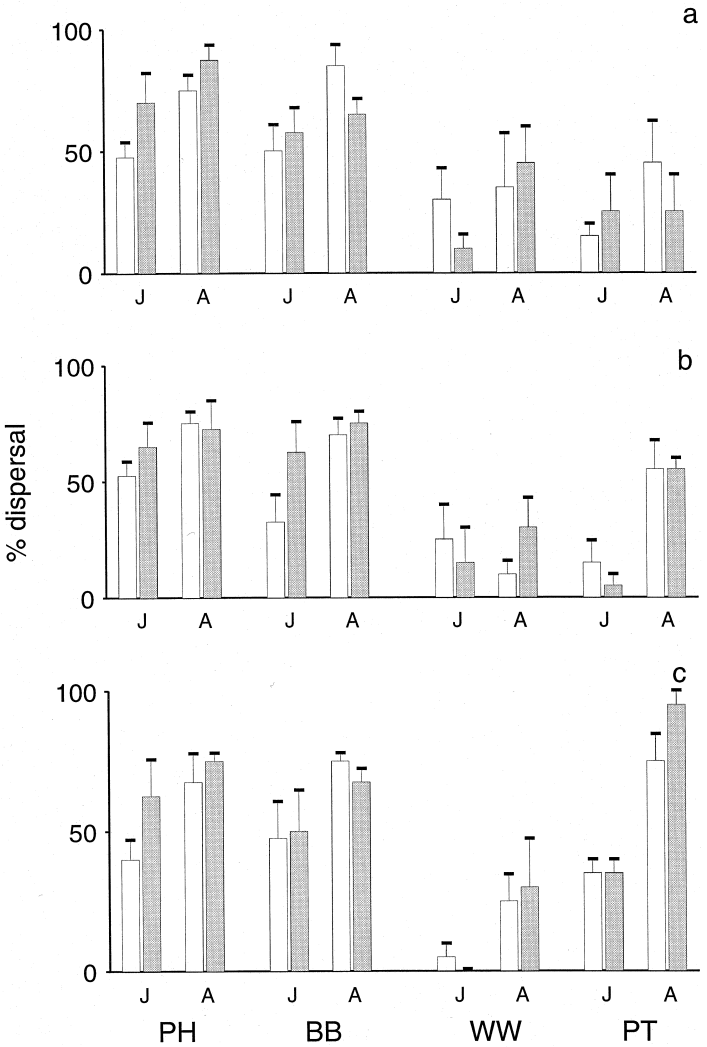

Fig. 4. Temporal variation in percentage dispersal of *Bembicium*. a. Summer 1993; b. Winter 1994; c. Winter 1995. Mean percentages of initial numbers dispersed + SE (n = number of plots = 4) are shown. Initial numbers were: 10 adults and 10 juveniles on rocky shores and 5 adults and 5 juveniles in mangrove forests. □ = plots contiguous with surrounding oysters; ▨ = plots isolated from surrounding oysters; J = juveniles; A = adults; Rocky shores: PH = Port Hacking, BB = Broken Bay; Mangrove forests: WW = Woolooware, PT = Patonga.

those distances were not as large on either of the subsequent runs of the experiment (SNK procedure, $P < 0.05$ and $P < 0.01$ respectively). The only spatial inconsistency detected for rocky shores was a difference in both the magnitude and the variance of

Table 5
Analyses of temporal variation in dispersal of adult and juvenile *Bembicium*. (a) Percentage dispersal ($n = 4$ plots; data are untransformed; Cochran's test: ns); (b) mean displaced distances ($n = 4$ plots; data are transformed $X' = \mathrm{Log}_e(X + 1)$; Cochran's test: ns); (c) variance within plots of displaced distances ($n = 4$ plots; data are transformed $X' = \mathrm{Log}_e(X + 1)$; Cochran's test: ns). Experimental treatments only. Initial densities were 5 juveniles and 5 adults per plot in mangrove forests and 10 juveniles and 10 adults per plot on rocky shores

Source	(a)			(b)			(c)		
	df	MS	*F*	df	MS	*F*	df	MS	*F*
Time = T(S)	2	331.77	0.14	2	2.16	0.58	2	3.57	0.26
Habitat = H	1	51679.69	No test	1	174.76	No test	1	572.93	No test
Location = L(H)	2	4263.02	1.79	2	5.08	1.37	2	29.10	2.12
Isolation = I	1	713.02	No test	1	1.23	No test	1	22.46	No test
Age = A	1	26367.19	No test	1	41.98	No test	1	168.62	No test
T × H	2	1135.94	0.48	2	3.50	0.94	2	2.08	0.15
T × L(H)	4	2388.02	5.40***	4	3.71	8.87***	4	13.74	[3]5.62***
T × I	2	150.52	0.32	2	0.70	0.58	2	1.34	0.18
T × A	2	432.81	0.65	2	0.003	0.00	2	2.35	1.30
H × I	1	713.02	No test	1	1.65	No test	1	5.45	No test
H × A	1	275.52	No test	1	5.26	No test	1	0.99	No test
I × L(H)[a]	2	275.52	0.58	2	0.76	0.63	2	2.33	0.31
A × L(H)[b]	2	1450.52	2.18	2	0.56	0.59	2	2.22	1.22
I × A	1	88.02	No test	1	2.23	No test	1	16.92	No test
T × H × I[c]	2	125.52	0.26	2	1.53	1.29	2	3.89	0.52
T × H × A[d]	2	441.15	0.66	2	0.91	0.95	2	1.14	0.63
I × T × L(H)[e]	4	475.52	1.08	4	1.19	[2]2.90*	4	7.41	[3]3.03*
A × T × L(H)[f]	4	666.15	1.51	4	0.96	[2]2.33	4	1.81	0.73
T × I × A[g]	2	134.90	0.40	2	0.002	0.02	2	1.78	0.79
H × I × A	1	2479.69	[1]5.74*	1	0.10	No test	1	3.79	[3]1.55
A × I × L(H)[h]	2	450.52	1.32	2	0.35	[2]0.86	2	1.60	0.71
T × H × I × A[i]	2	114.06	0.33	2	1.13	[2]2.78	2	4.25	1.89
A × I × T × L(H)[j]	4	341.15	0.77	4	0.11	0.27	4	2.25	0.91
Residual[k]	144	441.84		144	0.42		144	2.49	

*Denotes significance at $P < 0.05$; ** denotes significance at $P < 0.01$; *** denotes significance at $P < 0.001$.
[1]*F* calculated using pooled MS (a + c + e + g + h + i + j + k = 432.06 with 162 df).
[2]*F* calculated using pooled MS (g + j + k = 0.40 with 146 df).
[3]*F* calculated using pooled MS (b + d + f + g + h + i + j + k = 2.45 with 162 df).

displaced distances from contiguous plots between Port Hacking and Broken Bay in Summer 1993 (Table 5b; SNK procedure, $P < 0.01$ and Table 5c; SNK procedure, $P < 0.05$ respectively).

4. Discussion

In general, patterns of dispersal of *Bembicium* were remarkably consistent through time, particularly on rocky shores. The combined results of the experiments described above lend considerable weight to the conclusions drawn by Crowe (1996a) and suggest that the patterns described in that paper can be predicted reliably on rocky shores, but

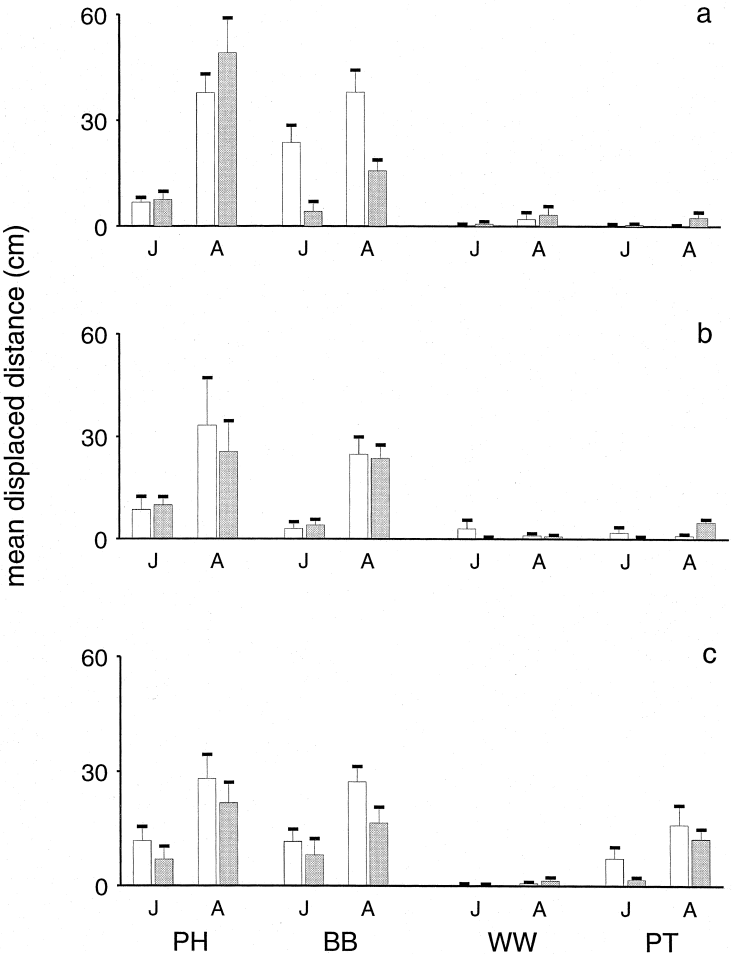

Fig. 5. Temporal variation in displaced distances of *Bembicium*. a. Summer 1993; b. Winter 1994; c. Winter 1995. Mean displaced distances + SE (n = number of plots = 4) are shown. The number of individual measurements contributing to the mean for any given plot varied from 3 to 10 and was generally smaller in mangrove forests than on rocky shores. Initial numbers were: 10 adults and 10 juveniles on rocky shores and 5 adults and 5 juveniles in mangrove forests. □ = plots contiguous with surrounding oysters; ▨ = plots isolated from surrounding oysters; J = juveniles; A = adults; Rocky shores: PH = Port Hacking, BB = Broken Bay; Mangrove forests: WW = Woolooware, PT = Patonga.

with somewhat less confidence for populations of *Bembicium* inhabiting mangrove forests. Implications of these patterns of dispersal for the population ecology of *Bembicium* are discussed in Crowe (1996a).

The model that consistent seasonal differences in behaviour of snails in mangrove forests caused the differences in dispersal observed in the first run of the dispersal experiment was not supported by the results of this test. Although a greater percentage

of snails at Patonga left plots during winter months than during summer months, there was no significant difference in displaced distances. Percentage dispersal is a more highly summarised measure of dispersal than displaced distance and may not always accurately reflect dispersive behaviour (Fig. 6). More weight should therefore be given to evidence from analyses of displaced distance when interpretation is not straightforward. The trend for increased percentage dispersal in winter at Patonga was reversed at Woolooware. It is therefore appropriate to use the evidence from the analyses of displaced distance to test the hypothesis critically. The null hypothesis was retained in both the analyses of mean displaced distances and within plot variances of displaced distances. Although results from controls are not presented here (see Section 2), they were done for each experiment and provided evidence that handling and disturbance did not significantly alter the dispersive behaviour of the snails.

The tests of hypotheses from the seasonal model were only evaluated in terms of the gross difference in rates of dispersal in the two habitats. The other major difference between the two habitats was that dispersal of juveniles was affected by the spatial structure of the microhabitat on rocky shores, but not in mangrove forests (Crowe, 1996a). Unfortunately, the seasonal comparison could only be done for adult snails, so the effect of season on this difference between the habitats could not be directly tested.

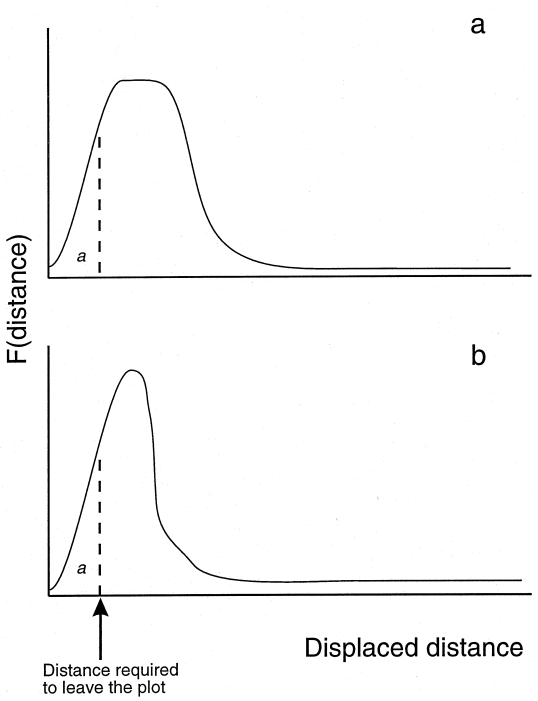

Fig. 6. Two frequency distributions (a and b) of displaced distances involving approximately equal numbers of individuals and having different means and variances (i.e. in which the behaviour of the two populations is different). The percentages of individuals that would be observed to have dispersed from a small plot would not be different: area *a* is the same in each case.

That pattern was, however, consistent through three temporal replicate runs of the dispersal experiment: one in summer and two in winter (see Section 3).

Although seasonality in dispersal was not found to be important here, seasonality in patterns of behaviour of marine gastropods and their impact on other members of local assemblages has been found to be more significant in other habitats and regions. For example, Underwood and Jernakoff (1984) demonstrated seasonal variations in the interaction between gastropods and algae on exposed shores in New South Wales and Cubit (1984) and Williams (1993) made similar observations in the United States and Hong Kong, respectively. Juvenile sea scallops in waters off Nova Scotia also showed seasonality in their patterns of movement (Carsen et al., 1995). The lack of seasonal pattern found in the current study is, however, by no means unique. Other communities have been shown to be relatively stable despite seasonal changes in the physical environment (e.g. Lubchenco et al., 1984; but see Ortega, 1987).

There were, however, several changes in behaviour through time that were unrelated to season. These differences were observed in both the seasonal and temporal comparisons, and were most prevalent in mangrove forests where the changes were sometimes quite dramatic. For example, there was greater dispersal at Patonga in Winter 1995 than was recorded there during other runs of the experiment. Such an increase was not observed at any other location during that run of the experiment. Thus, patterns of temporal variation in this process were inconsistent from place to place (cf. Kennelly and Underwood, 1993). The pattern of dramatic changes in behaviour of *Bembicium* at Woolooware observed by Underwood and Barrett (1990) were attributed by them to season, but the apparent association between winter and large rates of movement may have been coincidental and attributable instead to the sorts of irregular temporal changes observed here. Similarly, LaFuente (1996) recorded occasional aseasonal reproductive events in a species traditionally regarded as a seasonal breeder. It is clear from these results that comparisons of 'seasonal' patterns of behaviour made using only one observation during summer months and one during winter months can be quite misleading. Such unreplicated studies are still remarkably common (see, for example, Erlandsson and Kosteylev, 1995; Farnsworth and Ellison, 1996; Quiblierlloberas et al., 1996; Siguenza et al., 1996) despite recent papers (e.g. Hurlbert, 1984; Underwood, 1994a, Underwood, 1994b) refocussing attention on these problems.

Gross changes in behaviour observed at particular times of year are usually responses to particular cues that tend to coincide with that time of year. If such cues do not arise in any particular year, the response will not be elicited. To call such changes 'seasonal' is thus something of a misnomer, brought on by our anthropocentric view. As an illustration, Recher et al. (1996) described seasonal patterns of abundance in canopy-dwelling invertebrates in eucalypt forests. They went on, however, to point out that unseasonable rainfall can create unseasonable peaks in abundance. Conversely, Bell (1985) found that when there was no rain in the usual peak-season for invertebrates, the peak did not occur at all. Clearly, the animals are responding to rain and its effect on canopy foliage rather than to any seasonal calendar. Predictions of abundance based on rainfall would be more meaningful than those based on season.

This study has indicated the relatively consistent nature of dispersal on rocky shores and identified temporal and spatial variation in mangrove forests. Additional temporal

variation could occur at scales smaller or larger than those investigated in the current study. As described by Underwood (1991, Underwood, 1994a, Underwood, 1994b), variations on smaller time-scales can easily confound comparisons made at widely separated intervals. In some habitats, variation from week to week, or even from day to day, can outweigh that observed during monthly or quarterly sampling (e.g. Morrisey et al., 1992). Having identified and to some extent described variation through time, the next stage should be to investigate its causes, particularly in mangrove forests. An understanding of the causes would enable more detailed models to be developed and thus improve the precision of predictions. Although such work has not yet been done for *Bembicium*, the following discussion suggests some potentially fruitful avenues of research.

Movements of intertidal gastropods in relation to microhabitat have been found to vary in relation to environmental cues such as temperature and humidity (Feare, 1971; Lewis and Bowman, 1975; Menge, 1978; Moran, 1985; Burrows and Hughes, 1989). Variations in temperature, humidity or the timing of low tide could explain the differences in dispersal observed here. It is not clear, however, why such variations should affect snails in mangrove forests more than those on rocky shores. Intuitively, one would expect the mud and pneumatophores of the mangrove forest to buffer the effects of variations in temperature and humidity by casting shade and retaining moisture. Such protection is not present on rocky shores and yet dispersal was quite consistent through time.

Predators vary in abundance and activity at both regular and irregular intervals (e.g. Bernstein et al., 1981; Fairweather, 1988; Burrows and Hughes, 1989). Invertebrate prey can sense the presence of predators or damaged conspecifics and modify their behaviour as a result (Doering and Phillips, 1983; Forrester, 1994; Vadas et al., 1994). *Bembicium* could occasionally respond to the presence of particularly large numbers of predators, such as toadfish, by remaining closely associated with oysters, but disperse freely when numbers of toadfish are small. To test this model, it would be necessary (a) to document variation in abundance of predators and (b) to test the hypothesis that *Bembicium* alter their dispersive behaviour in the presence of predators, either in the laboratory or using enclosures in the field.

The fact that a great proportion of snails dispersed at Patonga in the Winter of 1995 coincided with the observation of a large number of mating snails, including many of the animals marked for the experiments (pers obs.). Changes in the migratory patterns of gastropods associated with reproduction have been documented by several authors (e.g. Kojima, 1959; Fretter and Graham, 1962; Feare, 1971). It is possible that reproductive activity caused the fast rates of movement of *Bembicium* at that time.

If availability of algal food is a factor influencing patterns of movement of *Bembicium*, then variations in the availability of that food could cause the variations in patterns of movement observed in mangrove forests. The relationship between patterns of movement of herbivores and the availability of their food has often been discussed (e.g. Hansson, 1977; Kareiva, 1982; Nagasaka, 1992; Lewis, 1994), but although Branch and Branch (1980) made preliminary investigations, the link between food and movement has not been thoroughly investigated for this species.

The comparative approach used here did not involve the methodology of meta-analysis, a technique that is increasingly used to assess the generality of ecological processes. Meta-analysis incorporates information on effect sizes in comparing the outcomes of disparate studies on the process with the aim of calculating its overall relative importance: a kind of 'quantitative review of the literature' (Gurevitch et al., 1992; Gurevitch and Hedges, 1993; Myers, 1995). The need for this technique arises from the fact that studies on similar processes are often unplanned and therefore use widely varying methods in very different situations and cannot be formally compared in any other way. While meta-analysis is a valuable tool for taking stock of the currently diverse literature, more valuable still are studies that involve the temporal repetition of spatially replicated experiments using the same methodology. Although Gurevitch and Hedges (1993) asserted that ''ecological experiments are essentially never replicated in any strict sense'', this is not and should not be the case (witness the current paper as well as studies of competition between two gastropods: Underwood, 1978, Underwood, 1984 (cited in Underwood and Petraitis, 1993). Studies must be designed so that the generality of the findings can be genuinely and quantitatively assessed. It is extremely unlikely that commensurable studies will arise in different geographic regions by chance (Underwood and Fairweather, 1986; Xia and Boonstra, 1992; Underwood and Petraitis, 1993). It will therefore be necessary for researchers either to initiate studies incorporating broad spatial and temporal replication or to collaborate with overseas workers to meet this challenge. Underwood and Petraitis (1993) and Beck (1997) have offered useful advice on appropriate designs for such comparisons.

It will not be possible to make proper geographic comparisons of ecological processes until we have some understanding of temporal variations in their action (Kennelly and Underwood, 1993; Underwood and Petraitis, 1993). If processes vary through time, as they have been demonstrated to do (e.g. Morrisey et al., 1992; Kennelly and Underwood, 1993; Haynes and Quinn, 1995, the current paper), then geographic comparisons based on a single observation at each location will be confounded by possible temporal differences. Experiments must therefore be replicated both spatially and temporally in order to assess the generality of the processes under investigation, both within and beyond the local region or habitat. In this way, it is more likely that urgently needed models capable of making accurate and precise predictions in a wide range of circumstances will emerge.

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